

I. INTRODUCTION

Data has become a strategic asset that drives innovation, strengthens public sector performance, and enables evidence-based policymaking. Recognizing this, governments worldwide have adopted enterprise data governance as a foundational capability for digital government, ensuring that data is accurate, trusted, secure, interoperable, and responsibly managed throughout its lifecycle.

The Department of Science and Technology (DOST) generates, manages, and relies on a uniquely valuable body of scientific, technical, and geospatial information across its Central Office, attached agencies, and regional offices. This information naturally led to the development of specialized information systems, data management practices, and institutional expertise. This distributed approach reflects the Department's strength as a network of scientific and technical institutions, each operating within distinct operational contexts, technical domains, and statutory responsibilities. As a result, DOST possesses a rich and diverse ecosystem of data assets that represent significant strategic value to both the Department and the nation.

At the same time, the decentralized evolution of data systems across the DOST System presents challenges in data discoverability, interoperability, accessibility, and reuse. Information that could support cross-cutting priorities, such as disaster resilience, environmental management, industrial competitiveness, scientific collaboration, and public service delivery, may be difficult to locate, integrate, validate, or utilize across organizational boundaries. These challenges also affect the Department's readiness to adopt advanced analytics, artificial intelligence, and other emerging technologies, whose effectiveness depends on the availability of discoverable, interoperable, well-governed, and trusted data assets.

Recognizing these realities, DOST envisions an organization that identifies data governance as a critical enabler of public sector innovation and integrated service delivery, emphasizing that effective governance transforms fragmented information into strategic national assets that improve institutional performance and public value.

The Department seeks to establish a DOST-Wide Data Governance Framework that will provide a strategic and enabling foundation for managing data as a shared institutional asset. The Framework does not seek to centralize ownership or control of agency data, nor does it seek to replace existing mandates, systems, or institutional expertise. Rather, it aims to establish common principles, shared governance structures, and collaborative mechanisms that strengthen trust, interoperability, and responsible data sharing across the DOST System while preserving the autonomy, stewardship, and specialized functions of individual agencies and offices.

The Data Governance Framework adopts a federated and collaborative approach to establishing a data enterprise that provides strategic direction through three interconnected governance layers: (1) the **Data Governance Council Layer**, which provides strategic leadership, policy direction, and oversight; (2) the **Data Stewardship Layer**, which ensures responsible data management, stewardship, and accountability across agencies and offices; and (3) the **Technical and Analytics Layer**, which supports the implementation, management, analysis, and secure use of data through appropriate technological and analytical capabilities. Together, these governance layers promote trusted data sharing, interoperability, innovation, and evidence-based decision-making while respecting the mandates, expertise, and autonomy of participating institutions. Through shared principles and coordinated governance

mechanisms, the Framework seeks to enhance the discoverability, quality, interoperability, usability, and strategic value of DOST data assets, thereby enabling evidence-based decision-making, fostering innovation, supporting artificial intelligence readiness, and strengthening the Department's contribution to national development.

As a strategic governance instrument, the Data Governance Framework provides a common governance structure that defines decision rights, accountability, stewardship, and guiding principles for data management across the DOST System. Supporting policies, standards, and procedures shall operationalize this Framework while allowing individual agencies to retain ownership and responsibility over their respective data assets.

II. OBJECTIVES

1. To define, approve, and communicate data strategies, policies and procedures;
2. To assign roles and delineate responsibilities to all personnel involved in data acquisition, and data management;
3. To understand and promote best practices in data management;
4. To ensure information and infrastructure security of agency's systems;
5. To manage and resolve data related concerns;
6. To provide guidance on planning and upgrade prioritization;
7. To ensure data quality, integrity, and interoperability across all systems and agencies;
8. Define clear roles, responsibilities, and decision rights for data management;and
9. Provide mechanisms for issue resolution, audit, and continuous improvement of data governance practices

III. CONTEXT

What is Data Governance?

Think of Data Governance as the DOST's guide for managing and protecting its valuable data assets.

Every day, the DOST generates and uses data to support its many functions—not only research and development, but also laboratory and testing services, weather and environmental monitoring, disaster risk reduction, health and innovation programs, administrative and financial operations, human resource management, procurement, regulatory activities, and other programs and services. This information is often stored in different places, such as laboratory notebooks, spreadsheets, databases, information systems, cloud platforms, agency servers, and official reports. When data is scattered across multiple locations, it becomes difficult to know which version is the most up to date, who owns it, who is authorized to access it, and who is responsible for maintaining it.

Data Governance provides a clear and consistent approach to managing data throughout its entire lifecycle. It helps ensure that all DOST data—whether scientific, technical, operational, administrative, financial, or institutional—is accurate, secure, reliable, and accessible to the right people when needed.

For DOST, this means putting in place common policies, standards, and clearly defined roles for how data is created, collected, processed, stored, shared, protected, retained, archived, and eventually disposed of across all Sectoral Councils, Research and Development Institutes (RDIs), Regional Offices, attached agencies, and other organizational units. By following these governance practices, DOST can make better use of its data, promote collaboration across the Department, improve operational efficiency, meet regulatory requirements, and support informed decision-making, research, innovation, and the delivery of quality public services.

Why is Data Governance Critical for DOST?

1. It Maximizes the Value of Public Funds

Every peso invested in the DOST—whether through research grants, scientific services, infrastructure, operations, or other programs—is expected to generate meaningful scientific and societal impact. However, the value of these investments can be reduced if the data used across the Department is inaccurate, duplicated, incomplete, inconsistent, or poorly managed.

This applies not only to research data, but to all DOST data, including those used in laboratory services, monitoring systems, administrative processes, regulatory functions, and day-to-day operations. When data is unreliable or stored in different formats and locations, even the most promising work can be delayed or compromised.

For instance, teams working on climate-resilient crops, disaster risk maps, public health analyses, or technology development all depend on high-quality and trustworthy data. If the underlying data is flawed, time and resources are often spent fixing errors instead of generating insights and solutions.

Data Governance helps prevent these issues by ensuring that all DOST data is accurate, consistent, and properly managed from the moment it is created. This reduces costly rework, improves efficiency across programs and services, and enables scientists, technical staff, and policymakers to make better decisions based on reliable and well-managed information.

2. It Supports Compliance with National Laws and Policies

As a government institution, DOST manages a wide range of sensitive information, including health records, research data, and community-level datasets. Protecting this information is both a legal responsibility and a public trust.

Data Governance provides the policies and processes needed to ensure that data is collected, stored, shared, and disposed of responsibly. It helps DOST comply with the Philippine Data Privacy Act of 2012 (RA 10173), the guidelines of the National Archives of the Philippines, and government open data initiatives.

By following these standards, DOST strengthens data security, supports transparency, and remains prepared for audits and regulatory requirements.

3. It Creates One Reliable Source of Truth

Many DOST programs involve collaboration among Sectoral Councils, Research and Development Institutes, and Regional Offices. Without common data standards, different offices may report different figures for the same project.

For example, one office may report that 15 technologies have been commercialized, while another reports only 10 completed licensing agreements. Instead of focusing on planning and innovation, teams spend time determining which numbers are correct.

Data Governance eliminates these inconsistencies by standardizing how data is defined, recorded, and reported across the entire DOST system. Whether monitoring project milestones, research outputs, budgets, or technology transfers, everyone works from the same verified and up-to-date information. This leads to better coordination, faster decision-making, and greater confidence in the data used across the organization.

IV. DEFINITION OF TERMS

Term	Definition	Reference
Data Sharing	The authorized, secure, and controlled exchange of data between DOST offices, attached agencies, external government entities, and approved stakeholders through established mechanisms, agreements, and security protocols to support interoperability, collaboration, analytics, and	RA 10173 (Data Privacy Act of 2012); DOST Data Privacy and Protection

	evidence-based decision-making while ensuring compliance with privacy and security regulations.	Policy Sec. 7; PeGIF
Data Governance Council	The highest governing and decision-making body within the DOST responsible for establishing strategic direction, approving policies and standards, resolving cross-agency data governance issues, overseeing compliance, and ensuring alignment of data governance initiatives with national ICT, privacy, and security regulations.	RA 10844 Sec. 6; DOST Data Governance Framework
Secure File Transfer Protocol (SFTP)	A secure network protocol used for transferring files between systems through encrypted communication channels and authenticated access mechanisms to ensure the confidentiality and integrity of data during transmission.	ISO/IEC 27002:2022; DOST Data Protection and Privacy Policy
Application Programming Interface (API)	A defined set of rules, protocols, and interfaces that enables different software applications, systems, or platforms to securely communicate, exchange data, and integrate functionalities in a standardized and interoperable manner.	PeGIF; ISO/IEC 27002:2022
Data Encryption	The process of converting readable data into encoded or ciphered information using cryptographic algorithms to prevent unauthorized access and protect the confidentiality and integrity of information both at rest and in transit.	ISO/IEC 27002:2022; RA 10173; DOST Data Privacy and Protection Policy
Data Retention	The process and policy governing the duration for which data and records are maintained, stored, and preserved based on legal, regulatory, operational, archival, and business requirements before archival or secure disposal.	RA 9470; DOST Digital Data Retention Policy; NPC IRR of RA 10173 Sec. 19(d)

Backup Data	Copies of critical information, databases, systems, applications, and records created and maintained for the purpose of restoration and recovery in the event of system failure, corruption, disaster, cyber incident, or accidental loss.	ISO/IEC 27002:2022 Control 8.13; DOST Backup and Recovery Policy
Recovery	The process of restoring data, systems, services, and operational capabilities from backup copies or recovery environments following data loss, system failure, disaster, or cybersecurity incidents to ensure continuity of operations.	DOST Backup and Recovery Policy; ISO/IEC 27002:2022
Acceptable Use Policy	A formal set of rules, responsibilities, and limitations governing the appropriate, authorized, ethical, and secure use of DOST information systems, networks, devices, internet resources, and data assets by employees, contractors, and third parties.	DOST Acceptable Use Policy; DOST ISMS Policy
Geospatial Data	Data that is associated with a specific geographic location or spatial reference and describes the physical features, boundaries, coordinates, or characteristics of natural or man-made objects, environments, and phenomena on Earth.	ISO 19157 Geographic Information — Data Quality; PeGIF

V. SCOPE

This framework applies to:

- All personnel (permanent, contracted, or third-party) within the DOST Central Office, Regional Offices, and Attached Agencies.
- All data types, encompassing both geospatial and non-geospatial data generated, collected, or managed by DOST entities.
- All technology assets used for data handling, including databases, cloud platforms, Application Programming Interfaces (APIs), web portals, and underlying hardware infrastructure.
- All stages of the data lifecycle: creation, acquisition, storage, processing, sharing, archiving, and disposal.

Implementation of this Framework shall align with:

- **DOST ICT Policies:** (Refer to specific DOST Administrative Orders or Circulars on ICT).
- **Republic Act No. 10173**, otherwise known as the **Data Privacy Act of 2012 (DPA)**, and its Implementing Rules and Regulations (IRR).
- **National Privacy Commission (NPC)** issuances, including Circulars, Advisory Opinions, and other guidelines.
- **Republic Act No. 9470**, otherwise known as the **National Archives of the Philippines Act of 2007**, and its IRR.
- Relevant International Organization for Standardization (ISO) standards, particularly **ISO/IEC 27001** (Information Security Management), **ISO 19157 (Geographic Information - Data Quality)**, and **ISO/IEC 25012 (Software engineering - Systems and software Quality Requirements and Evaluation (SQuaRE) - Data quality model)**.
- Other applicable national and international government regulations and industry best practices.

In cases of overlap or conflict among standards and regulations, this Framework shall always adopt the most stringent applicable standard to ensure the highest level of data protection and quality.

VI. FRAMEWORK: THE DOST DATA HOUSE

The framework adopts a multi-layered governance model to ensure strategic alignment and technical execution.

1. Data Governance Council Layer

The central governing body, providing high-level strategic direction, policy formulation, and regulatory oversight for data governance initiatives across the DOST System.

2. Data Stewardship Layer

The operational execution layer responsible for the direct management, compliance, and day-to-day quality control of DOST data assets. It consists of:

- a. **Data Owners:** Executive-level leadership (Agency Heads/Division Chiefs) accountable for data strategies, domain classification, and resource allocation.

- b. **Data Stewards:** Domain subject matter experts (Project Leaders/Researchers) who drive data quality, standardization, lifecycle management, and metadata accuracy.
- c. **Data Custodians (IT Role):** Systems and network personnel responsible for infrastructure security, platform controls, and disaster recovery.

3. Technical and Analytics Layer

This layer implements data governance through technical systems, data processing, and analytical capabilities. It consists of:

- a. **Analytics Subgroup:** Focuses on leveraging data for insights.
 - i. **Analytics Manager:** Aligns analytics initiatives with strategic organizational goals.
 - ii. **Data Architect:** Designs the enterprise data architecture, data models, and data flow processes.
 - iii. **Data Engineers:** Develop and maintain Extract, Transform, Load (ETL) pipelines, data integration processes, and data warehousing solutions.
 - iv. **Data Analysts:** Perform data analysis, create visualizations, and generate reports to support decision-making.
 - v. **Data Collection Officers:** Coordinate and validate data acquisition from various sources.
- b. **Technology Subgroup:** Ensures the underlying technical infrastructure is robust and secure.
 - i. **IT Systems and Database Specialists:** Maintain server infrastructure, ensure system reliability, implement security controls, and manage database systems.
 - ii. **Database End-users:** Authorized personnel who access and utilize data directly for operational purposes.

VII. ROLES AND RESPONSIBILITIES

A. DATA GOVERNANCE COUNCIL

- **Policy Formulation:** Formulate, approve, and communicate high-level departmental data strategies, governance policies, and technical procedures across all DOST bureaus and regional offices.

- **Role Delineation:** Define clear roles, responsibilities, and decision rights for data management personnel involved in data acquisition and long-term storage.
- **Conflict Resolution:** Act as the ultimate mechanism for resolving internal data-related concerns, cross-agency data ownership disputes, and usage conflicts.
- **National Compliance:** Monitor compliance with data governance policies, approve data-sharing agreements, and elevate unresolved inter-agency issues to the Department of Information and Communications Technology (DICT) per RA 10844 Section 6.
- **Continuous Improvement:** Provide overarching frameworks for audits, key performance evaluations, and the continuous improvement of data governance practices across the department.

B. DATA STEWARDSHIP LAYER

a. Data Owners (Agency Heads/Division Chiefs)

- **Strategic Alignment:** Approve domain-specific data strategies and translate Council policies into localized operational directives.
- **Accountability:** Take ultimate accountability for the business value, legal compliance, and data classification (public, restricted, confidential) within their respective domains.
- **Lifecycle Management:** Oversee the complete lifecycle of their domain's data assets from initial acquisition to archiving or final disposal.
- **Planning & Prioritization:** Review operational data gaps to provide strategic guidance on planning and upgrade prioritization for agency systems and data infrastructure.

b. Data Stewards (Project Leaders / Senior Researchers)

- **Data Quality & Integrity:** Enforce data quality rules, conduct data cleansing activities, and directly resolve day-to-day data-related quality issues to ensure data integrity.
- **Standards & Best Practices:** Promote best practices in data management within project teams, ensuring meticulous metadata documentation and clear data lineage tracking.
- **Interoperability:** Ensure that project-level datasets adhere to standard schemas and vocabulary to support cross-agency interoperability.
- **Audit Support:** Serve as the primary point of contact for operational data audits and provide technical inputs to help resolve localized data usage concerns.

c. Data Custodians (IT Division / Systems Administrators)

- **Infrastructure Security:** Implement and enforce strict technical, logical, and physical access controls, encryption protocols, security patches, and firewalls to ensure the infrastructure security of the agency's systems.
- **Technical Solutions:** Build and maintain secure data storage platforms, including localized databases, central data lakes, and secure cloud storage configurations.
- **Data Availability:** Execute and test routine data backup, business continuity, and disaster recovery procedures to maximize systemic reliability and uptime.
- **System Upgrades:** Maintain the hardware, servers, and networks supporting data assets, offering system-level feedback on necessary technology upgrades.

C. TECHNICAL AND ANALYTICS SUBGROUP

a. Analytics Subgroup

- **Analytics Manager**
 1. **Strategic Mapping:** Align analytics initiatives and data science projects with the strategic organizational goals, mandates, and performance metrics of the DOST.
 2. **Upgrade Guidance:** Monitor analytical tool trends and process constraints to provide guidance on analytical platform planning and upgrade prioritization.
- **Data Architect**
 1. **Interoperability Design:** Design the enterprise data architecture, data models, schemas, and blueprint flow processes specifically tailored to ensure data integrity and seamless interoperability across all DOST systems and agencies.
 2. **Best Practice Standards:** Define the blueprint requirements for data schemas, ensuring data integration processes naturally follow industry best practices.
- **Data Engineers**
 1. **Pipeline Execution:** Build, optimize, and maintain reliable Extract, Transform, Load (ETL) pipelines, data integration processes, and data warehousing solutions.
 2. **Continuous Improvement:** Programmatically implement mechanisms for data verification, error handling, and continuous improvement of automated data processing pipelines.
- **Data Analysts**
 1. **Insights Extraction:** Perform deep data analysis, build interactive dashboards, and generate reports from data to support leadership in strategic decision-making.

2. **Audit Reporting:** Analyze data access logs, metrics discrepancies, and system outputs to provide descriptive reports for governance compliance reviews.

- **Data Collection Officers**

1. **Data Acquisition Compliance:** Coordinate and validate raw data acquisition from various regional and institutional sources, ensuring personnel follow assigned collection protocols.
2. **Source Quality:** Intercept and address raw data quality or missing attribute anomalies at the point of ingestion before data enters the engineering pipelines.

- b. **Data Infrastructure Subgroup**

- **IT Systems and Database Specialists**

1. **System Integrity:** Provide deep technical expertise for database management systems (DBMS), server configuration, system maintenance, and performance tuning to maintain system reliability.
2. **Security Enforcement:** Implement technical controls, user privilege management, and access monitoring tools as directed by the Data Custodian to protect data environment security.

- **Database End Users (Authorized Personnel / Researchers)**

1. **Policy Adherence:** Access and utilize DOST data assets directly for approved operational, research, or administrative purposes while strictly adhering to the agency's "**Acceptable Use Policy.**"
2. **Issue Reporting:** Understand and actively follow standard reporting protocols to flag data inconsistencies, system anomalies, or security risks to Data Stewards or IT personnel.

VIII. GOVERNANCE COMPONENTS

A. Data Discovery and Data Inventory

Effective data protection and governance rely on a thorough understanding of the organization's information assets.

1. All DOST units and offices must systematically identify, record, and regularly update a comprehensive inventory of all data and documents they manage.
2. The Data Inventory shall serve as the centralized data catalog, documenting:

- a) Data sources and origin;
 - b) Assigned Data owners and stewards;
 - c) Data classification;
 - d) Retention Requirements;
 - e) Format;and
 - f) Data Collection strategy.
3. The inventory must be continuously updated and auditable to support compliance and governance objectives, aligning with **ISO/IEC 27002:2022, Control 5.9 (Inventory of information and other associated assets)**.

B. Data Classification

After identifying data assets and inventory that will be administered, all data are subject for data classification and it should be in accordance with the existing DOST Data Classification under Data Protection and Privacy Policy:

Level	Access	Description	Examples
Public	Unrestricted access; available to the general public.	Information that can be freely shared with the public without any restrictions.	Press releases, published research reports, public announcements, etc.
Restricted (Internal Data)	Limited to DOST personnel or relevant stakeholders.	Information intended for internal use within DOST, its agencies and regional offices, and affected third parties but not publicly available.	Internal memos, special orders, administrative orders, operational procedures, employee directories, etc.
Confidential	Authorized personnel on a need-to-know basis.	Sensitive information that should be accessed only by authorized	Personnel records, Network Diagrams, Project Proposals, and Procurement

		personnel due to privacy, legal, or security considerations.	documents, etc.
Secret	Strictly limited to select personnel with special clearance.	Highly sensitive information requiring strict access control and encryption. Unauthorized access could lead to significant legal, financial, reputational, operational, or national security risks.	Confidential DOST Project Proposals, Funding Allocations, Government Contracts, Risk Assessment Reports, etc.
Top Secret	Extremely restricted access with highest security clearance.	Information or material that, if disclosed, could cause exceptionally grave damage to national security.	Classified national security research (e.g., defense tech, cybersecurity strategies)

C. Data Acquisition

1. The DOST shall implement standardized and secure data acquisition processes to support interoperability, data quality, accountability, and evidence-based decision making across the DOST Central Office, Regional Offices, and all Attached agencies. All data acquisition and transfer activities shall comply with the RA 10173 (Data Privacy Act of 2012), the DOST Data Protection and Privacy Policy, and all applicable information security and governance standards. *(Ref DOST Data Protection and Privacy Policy Sec. 1; Sec. 2; RA 10173).*
2. The DOST will employ secure platforms for data transfer, such as **Secure File Transfer Protocol (SFTP)** or web portal services (e.g., **RESTful APIs**), to ensure seamless and secure transfer of files. All data transfers will adhere to established security protocols, including encryption (e.g., TLS/SSL) and user authentication, to maintain the confidentiality and

integrity of the data, consistent with ISO/IEC 27002:2022, Control 8.24 (Secure coding) and Control 8.22 (Development security requirements).

3. Data collection and processing activities shall adhere to the principles of lawful, purposeful, transparent, and authorized processing, ensuring that data is collected only for legitimate organizational purposes and accessed solely by authorized personnel in accordance with assigned roles and responsibilities. (Ref. *DOST Data Protection and Privacy Policy, Sec. 4; RA 10173*).
4. Data Owners, Data Stewards, Data Custodians, and designated agency focal persons throughout the DOST System shall coordinate data submission, validation, integration, and monitoring activities to ensure data quality, consistency, accountability, and auditability across the data lifecycle. (Ref. *ISO/IEC 27002:2022 Control 8.15 – Logging; DAMA DMBOK*).

D. Data Quality Management

To ensure that all information assets across the Department and its attached agencies are reliable and AI-ready, the DOST-wide Data Governance Framework (DW-DGF) mandates a standardized approach to data health. Data quality is paramount to ensuring that collected datasets are reliable, usable, and fit for their intended purpose.

1. Data Standardization and Cleansing

All data ingested from various DOST agencies must undergo a standardized cleansing process to eliminate duplicates, correct errors, and harmonize formats. This ensures interoperability between disparate systems (e.g., merging regional geospatial data with central administrative records).

2. Applicable International Standards

Data quality ensures that the collected datasets are reliable, usable, and fit for their intended purpose. It shall adopt standards such as:

- a. ISO 19157 (Geospatial)
- b. ISO 25012 (General Data Quality)

3. Measurable Key Performance Indicators (KPIs)

Include measurable **KPIs**:

- Accuracy Rate: Percentage of data records free from errors.

- **Completeness Percentage:** Percentage of required data fields that are populated.
- **Timeliness:** Lag time between data event and its availability in the system.
- **Consistency:** Degree to which data values are uniform across different systems or over time.
- **Validity:** Conformity of data to defined formats, types, and ranges.

(Ref: *ISO/IEC 25012 sec. 5.3. Data Characteristics*)

E. Data Architecture and Storage

DOST's data architecture and storage solutions will be designed to support high availability, scalability, security, and data sovereignty.

- **Architecture Model:** Define whether a centralized data warehouse/lake architecture will be implemented, or a federated/distributed model, as appropriate for different data types and agency needs.
- **Data Sovereignty:** Ensure that data classified as "Secret" or "Top Secret," and certain categories of "Confidential" and "Sensitive Personal Information" under **RA 10173**, are hosted within Philippine territory where legally required or operationally critical whether it is cloud based or on-prem.
- **Data Warehouse/Lake/Lakehouse:** Implement solutions for large-scale data storage and analytics.
- **Backup Systems:** Employ a 3-2-1 backup strategy (three copies of data, on two different media, with one copy offsite) to ensure resilience against data loss, aligning with **ISO/IEC 27002:2022, Control 8.13 (Information backup)** stating that backup copies of information, software, and systems, should be maintained and regularly tested in accordance with the agreed topic-specific policy on backup.
- **High Availability and Failover Systems:** Implement technologies (e.g., clustering, replication) to ensure continuous data access and minimize downtime, addressing requirements for critical operational data.

F. Data Access and Control

The DOST shall adopt and align its access and control mechanisms with the agency's existing Access and Control Policy. This ensures that our controls meet national standards for protecting the confidentiality, integrity, and availability of all information assets.

Physical Access Controls

- Sensitive facilities must have clearly defined security perimeters with physical barriers, entry controls, alarms, and continuous monitoring.

- Only authorized personnel may enter secure areas; visitor access must be escorted and logged.
- Access rights to physical areas must be reviewed regularly and revoked promptly when no longer needed.

Logical Access Controls

- Least Privilege Principle: Users and systems are given only the access strictly necessary to perform their roles.
- Need-to-Know and Need-to-Use Principles: Access to sensitive information is restricted to specific tasks and infrastructure where there is a documented requirement.
- Just-In-Time (JIT) Access: Temporary access is granted only for the period needed and is automatically revoked afterward.
- Identity and Authentication Management: Unique user identities must be verified, managed throughout their lifecycle, and promptly disabled when roles change or personnel leave.
- Password and MFA Requirements: Strong password policies, multifactor authentication, and prohibition of password reuse must be enforced.

Segregation of Duties

- Duties must be separated to prevent a single individual from performing conflicting roles (e.g., requesting and approving access).
- Access rights provisioning and approval must be handled by different roles to avoid abuse.
- Provisioning and Revocation of Access
- Access to information assets must be explicitly authorized by the asset owner.
- Temporary access rights should have expiration dates and be automatically removed.
- All access rights must be promptly revoked when no longer needed (e.g., upon resignation or role change). (*DOST Access Control Policy*).

G. Data Privacy and Protection

To safeguard personal and sensitive data, the DOST shall strictly adhere to **Republic Act No. 10173 (Data Privacy Act of 2012)**, its Implementing Rules and Regulations (IRR), and guidelines issued by the **National Privacy Commission (NPC)**. This also aligns with **ISO/IEC 27701 (Privacy Information Management System)**.

1. Data Collection & Processing Principles:

- a) **LAWFUL:** All data collection activities must comply with applicable laws and regulations.
- b) **PURPOSEFUL:** Data should only be collected for specific, legitimate purposes and not processed beyond the stated purposes.

- c) **CONSENSUAL:** Where required, data must be collected with the informed consent of the data subject.

Processing of personal and confidential data shall be restricted to authorized personnel who have legitimate access based on their roles and responsibilities

Data subjects are entitled to:

- a) Access their personal information upon request.
- b) Correct any inaccuracies in their data.
- c) Request deletion of their data, subject to legal and regulatory conditions.

(Ref. DOST Data Privacy and Protection Sec. 4)

- 2. Data Classification: Application of the **DOST Data Classification (Section B)** to personal and sensitive personal information, with corresponding security measures for each level.

- 3. Security Measures:

- a) Access Control – Data access is restricted based on job roles and responsibilities.
- b) Encryption – Sensitive data must be encrypted at rest and in transit.
- c) Secure Storage – Physical and digital storage locations must be protected against unauthorized access.
- d) Regular Audits – Periodic security assessments are conducted to ensure compliance.
- e) Incident Response – Any data breach must be reported immediately to the designated Data Protection Officer (DPO) and managed as per the incident response plan.
- f) Privacy and security measures must be integrated and evident throughout all phases of the System Development Lifecycle (SDLC).

(Ref. DOST Data Privacy and Protection Sec. 6)

H. Data Sharing and Integration

The DOST-Wide Data Governance Framework establishes standardized, secure, and controlled mechanisms for data sharing and integration across the DOST System and with authorized external stakeholders, ensuring compliance with applicable laws, privacy requirements, and information security standards. *(DOST Data Protection and Privacy Policy, Sec. 7; RA 10173).*

- 1. **Authorized Pathways:** Define approved mechanisms for data sharing (e.g., secure APIs, secure data transfer protocols like SFTP, restricted access portals).

2. **Standardized Metadata:** Utilize standardized metadata to improve data discovery, understanding, and interoperability across shared datasets, consistent with ISO 19115 (Geographic Information - Metadata) for geospatial data.

I. Data Retention, Archiving, and Disposal

The DOST adopts and strictly complies with the DOST Digital Data Retention Policy and Republic Act No. 9470 (National Archives of the Philippines Act of 2007) to ensure that all data assets are retained, archived, and disposed of in a secure, consistent, and auditable manner.

1. Retention Principles:

- a) Data shall be retained based on its classification, purpose, and relevant legal, regulatory, operational, and business requirements.
- b) Retention periods shall commence after the data is no longer actively in use for its primary operational purpose.
- c) Only data necessary for operational, legal, audit, or historical purposes shall be retained.
- d) Sensitive and personal data shall be retained only for as long as necessary and strictly in accordance with **RA 10173 and NPC issuances**. (*Ref. IRR of RA 10173 Section 19(d)*).
- e) Data Owners are accountable for ensuring compliance with prescribed retention periods.

2. Archiving Principles:

- a) Archiving solutions must ensure the long-term preservation, integrity, and retrievability of data deemed to have enduring value.
- b) Archived data must be securely stored, protected from unauthorized access or alteration, and regularly verified for integrity.
- c) **Physical Records Archiving (RA 9470):** All physical records with enduring value, as determined by the DOST and in accordance with **Republic Act No. 9470, Sections 2, 15, 18, and 20**, shall be transferred to a permanent government repository (e.g., National Archives of the Philippines) for proper management, control, and regulation of record disposition. The DOST shall regularly conduct an inventory of its physical records, maintain a registry, and ensure transparent procedures for their transfer, retention, and disposal. No physical record shall be disposed of or destroyed without the prior written authority of the National Archives.

3. Disposal Principles:

- a) **Secure Deletion Methods:** Employ industry-standard secure deletion methods, such as data wiping for digital media or physical destruction (e.g., shredding, degaussing) for storage devices that cannot be securely wiped. This aligns with **ISO/IEC 27002:2022, Control 7.10 (Disposal of media) Secure reuse or disposal**.
- b) **Documentation:** All data disposal activities must be logged, including the method used, date, and approving Data Owner. **(Ref. NPC IRR of RA 10173 Sec. 26(c))**

J. Data Backup and Recovery

The DOST shall adopt and implement the DOST Backup and Recovery Policy to ensure the protection, integrity, availability, and recoverability of all critical information assets and systems. Backup and recovery controls shall align with the DOST ISMS Policy, ISO/IEC 27002:2022 Control 8.13 (Information Backup), and applicable government regulations and standards. (DOST Backup and Recovery Policy, Sec. 1; Sec. 2; ISO/IEC 27002:2022)

1. Data Backup Standards

All critical data, systems, software, operating systems, control files, and supporting materials essential to DOST operations shall be identified and backed up based on operational and business requirements. Backup procedures, schedules, retention periods, storage locations, and recovery processes must be documented and regularly reviewed. (DOST Backup and Recovery Policy, Sec. 3; Sec. 4).

2. Backup and Recovery Strategy

The DOST shall implement a resilient backup strategy that includes:

- a) Weekly full backups and daily differential or incremental backups.
- b) Off-site backup storage copies for disaster recovery purposes.
- c) Regular recovery and restoration testing; and
- d) Secure disposal of backup media containing confidential information.
(DOST Backup and Recovery Policy, Sec. 5; Sec. 9; Sec. 11; Sec. 12)

The DOST may adopt the **high-availability failover systems** with the **3-2-1 backup principle**, maintaining multiple copies of data across separate storage media with at least one copy stored off-site to ensure resilience against system failures, cyber incidents, and disasters to implement a robust data protection approach. (ISO/IEC 27002:2022; DOST Backup and Recovery Policy, Sec. 9).

K. IT Data Management

The IT team plays a crucial role in data governance by providing the foundational technical capabilities for encryption, auditing, security, and infrastructure management essential for the successful implementation and continuous operation of this Framework.

1. **Technical Implementation:** Implement and manage technical controls for data access, security, and storage.
2. **Monitoring and Auditing:** Deploy tools and processes for continuous monitoring of data access, system logs, and security events, ensuring auditing capabilities in compliance with **RA 10173 and ISO/IEC 27002:2022, Control 8.16 (Monitoring activities)**.
3. **Infrastructure Management:** Ensure the stability, performance, and security of all data-related infrastructure, including networks, servers, and storage.
4. **Support & Training:** Provide technical support and training to Data Owners, Data Stewards, and end-users on data management tools, security protocols, and governance processes.

IX. Conclusion

In conclusion, data governance is essential to the **DOST** as its operations, services, research, and decision-making processes become increasingly data-driven. Establishing a Data Governance Framework provides DOST with a structured, consistent, and strategic approach to managing data as a valuable organizational asset. The framework aims to support all DOST offices, attached agencies, and regional operations by promoting accountability, data quality, security, interoperability, and responsible data sharing while remaining flexible enough to address the unique operational requirements of each unit.

Furthermore, the DOST Data Governance Framework shall serve as a living document that will continuously evolve alongside emerging technologies, policies, organizational needs, and data governance best practices. The framework shall be periodically reviewed and updated to ensure its relevance, effectiveness, and alignment with national government digital transformation initiatives, cybersecurity standards, and data management objectives. Through this framework, DOST strengthens its commitment to evidence-based governance, innovation, transparency, and the responsible stewardship of government data assets.

X. References and Regulatory Alignments

- *Acceptable Use Policy*
- *Data Management Body of Knowledge (DAMA-DMBOK)*
- *Data Privacy Act of 2012 (RA 10173)*
- *DOST Access Control Policy*

- *DOST Backup and Recovery Policy*
- *DOST Data Privacy and Protection Policy*
- *DOST Digital Data Retention Policy*
- *DOST ISMS Policy*
- *ISO/IEC 27002: 2022 Information security, cybersecurity and privacy protection — Information security controls*
- *ISO 19157 Geographic Information — Data Quality Standard*
- *ISO 25012 Software Engineering — Software Product Quality Requirements and Evaluation (SQuaRE) — Data Quality Model*
- *NPC IRR of RA 10173 - Implementing Rules and Regulations of the Data Privacy Act of 2012*
- *Philippine eGovernment Interoperability Framework (PeGIF)*
- *RA 10844 - An Act Establishing the Department of Information and Communications Technology (DICT)*
- *RA 9470 (National Archives of the Philippines Act of 2007)*